MA087G Mathematical Statistics
Beginner-Friendly Full Solution Guide
2025-08-23

Source notes before solving

The official exam PDF was image-only. That means normal text extraction found no text, but the rendered page images were readable.

One small source warning:

- In Uppgift 3c, the printed expression is visually a little unclear. The sensible reading, also used by student attempts, is $Y = X_2 - 2X_1$. I solve that version. If the expression were read literally as $Y = X_2 - 2X_2 = -X_2$, then $E(Y) = -2$ still, but $V(Y) = 2$ instead of 10.

Uppgift 1

What this question is really asking

We have two events, A and B. An event just means something that may or may not happen.

The question gives probabilities involving A^c and B^c. The little c means "not". So:

$$A^c = \text{not } A, \qquad B^c = \text{not } B.$$

We must find $P(A)$ and $P(B)$, meaning the probability that A happens and the probability that B happens.

How to recognize this type of question

The clues are:

- The question contains events A, B, A^c, and B^c.
- It gives conditional probabilities like $P(A^c \mid B^c)$.
- It gives a union probability $P(A^c \cup B^c)$.

So this is a probability algebra question. We should translate each line into formulas and solve for the missing probabilities.

Tiny theory needed

The symbol $P(A^c \mid B^c)$ means:

probability that A does not happen, given that B does not happen.

The formula for conditional probability is:

$$P(A^c \mid B^c) = \frac{P(A^c \cap B^c)}{P(B^c)}.$$

The symbol $\cap$ means "and". So $A^c \cap B^c$ means "not A and not B".

The symbol $\cup$ means "or". The formula for an "or" probability is:

$$P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c \cap B^c).$$

We subtract the overlap once because otherwise it gets counted twice.

Formula and what everything means

We will use:

$$P(A^c \mid B^c) = \frac{P(A^c \cap B^c)}{P(B^c)}$$

and

$$P(B^c \mid A^c) = \frac{P(A^c \cap B^c)}{P(A^c)}.$$

Here:

- $P(A^c \cap B^c)$ is the probability that neither A nor B happens.
- $P(A^c)$ is the probability that A does not happen.
- $P(B^c)$ is the probability that B does not happen.
- $P(A) = 1 - P(A^c)$, because A either happens or does not happen.
- $P(B) = 1 - P(B^c)$, for the same reason.

Step-by-step solution

Let

$$x = P(A^c \cap B^c).$$

This is just a shortcut so we do not have to write the long expression all the time.

The first given probability is:

$$P(A^c \mid B^c) = 0.6.$$

Use the conditional probability formula:

$$\begin{aligned} 0.6 &= \frac{P(A^c \cap B^c)}{P(B^c)} \\ &= \frac{x}{P(B^c)}. \end{aligned}$$

This says x is 60% of $P(B^c)$. Solve for $P(B^c)$:

$$\begin{aligned} 0.6P(B^c) &= x \\ P(B^c) &= \frac{x}{0.6} = \frac{5x}{3}. \end{aligned}$$

The second given probability is:

$$P(B^c \mid A^c) = 0.4.$$

Again use conditional probability:

$$\begin{aligned} 0.4 &= \frac{P(A^c \cap B^c)}{P(A^c)} \\ &= \frac{x}{P(A^c)}. \end{aligned}$$

Now solve for $P(A^c)$:

$$\begin{aligned} 0.4P(A^c) &= x \\ P(A^c) &= \frac{x}{0.4} = \frac{5x}{2}. \end{aligned}$$

The third given probability is:

$$P(A^c \cup B^c) = 0.8.$$

Now use the union formula:

$$P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c \cap B^c).$$

Insert what we know:

$$0.8 = \frac{5x}{2} + \frac{5x}{3} - x.$$

Now put everything over the same denominator, 6:

$$\begin{aligned} 0.8 &= \frac{15x}{6} + \frac{10x}{6} - \frac{6x}{6} \\ &= \frac{19x}{6}. \end{aligned}$$

Solve for x:

$$\begin{aligned} x &= 0.8 \cdot \frac{6}{19} \\ &= \frac{4}{5} \cdot \frac{6}{19} \\ &= \frac{24}{95}. \end{aligned}$$

Now find $P(A^c)$:

$$\begin{aligned} P(A^c) &= \frac{5x}{2} \\ &= \frac{5}{2} \cdot \frac{24}{95} \\ &= \frac{60}{95} = \frac{12}{19}. \end{aligned}$$

So

$$\begin{aligned} P(A) &= 1 - P(A^c) \\ &= 1 - \frac{12}{19} \\ &= \frac{7}{19}. \end{aligned}$$

Now find $P(B^c)$:

$$\begin{aligned} P(B^c) &= \frac{5x}{3} \\ &= \frac{5}{3} \cdot \frac{24}{95} \\ &= \frac{40}{95} = \frac{8}{19}. \end{aligned}$$

So

$$\begin{aligned} P(B) &= 1 - P(B^c) \\ &= 1 - \frac{8}{19} \\ &= \frac{11}{19}. \end{aligned}$$

Answer

$$\boxed{P(A) = \frac{7}{19} \approx 0.368}$$

$$\boxed{P(B) = \frac{11}{19} \approx 0.579}$$

What to memorize for the exam

When you see conditional probabilities with complements, introduce a small unknown for the overlap, for example:

$$x = P(A^c \cap B^c).$$

Then rewrite the conditional probabilities using fractions.

Common trap

Do not think $P(A^c \mid B^c)$ means $P(A^c) \cdot P(B^c)$. Conditional probability is a fraction, not multiplication.

Repeated exam pattern

This is the same basic pattern as 2026-01, Uppgift 1: conditional probability with complements. The numbers and wording changed, but the method is the same: translate the probability statements, use conditional probability formulas, and solve for the missing probabilities.

Uppgift 2

What this question is really asking

The question gives a function $f(x)$. We must make it into a probability density function.

A density function is used for a continuous random variable, meaning a variable that can take values on an interval, not just separate values like $0, 1, 2$.

We need to:

- find a, the missing constant,
- find the distribution function $F_X(x)$,
- calculate $P(X > 1)$,
- calculate expected value and standard deviation.

How to recognize this type of question

The clues are:

- The question says "täthetsfunktion", which means density function.
- The function is given on an interval $0 < x < 3$.
- It asks for F_X, which is the distribution function/CDF.
- It asks for expected value and standard deviation.

So we should use integrals. For continuous variables, sums become integrals.

Tiny theory needed

The density is:

$$f(x) = \begin{cases} ax^2, & 0 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

For f to be a density, the total area under the curve must be 1. That is because total probability is always 1.

So we need:

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

But $f(x) = 0$ outside $0 < x < 3$, so we only integrate from 0 to 3.

Formula and what everything means

Density condition:

$$\int_0^3 f(x)\,dx = 1.$$

Here:

- $f(x)$ is the density.
- The integral means area under the curve.
- The answer must be 1, because all probabilities together add to 1.

Distribution function:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt.$$

Here:

- $F_X(x)$ means probability that X is at most x.
- t is just a dummy integration letter. It plays the same role as x, but avoids confusion.

Expected value:

$$E(X) = \int x f(x)\,dx.$$

This is the average value of X in the long run.

Variance:

$$V(X) = E(X^2) - [E(X)]^2.$$

Standard deviation:

$$D(X) = \sqrt{V(X)}.$$

The standard deviation tells us roughly how spread out X is.

Step-by-step solution

Part a: Find a.

Use the density condition:

$$\int_0^3 ax^2\,dx = 1.$$

The constant a can be moved outside the integral:

$$a\int_0^3 x^2\,dx = 1.$$

Now integrate x^2:

$$\int x^2\,dx = \frac{x^3}{3}.$$

So:

$$\begin{aligned} a\left[\frac{x^3}{3}\right]_0^3 &= 1 \\ a\left(\frac{3^3}{3} - \frac{0^3}{3}\right) &= 1 \\ a\left(\frac{27}{3}\right) &= 1 \\ 9a &= 1 \\ a &= \frac{1}{9}. \end{aligned}$$

Part b: Find $F_X(x)$.

There are three regions:

- If $x \leq 0$, then we are before the density starts, so $F_X(x) = 0$.
- If $0 < x < 3$, integrate from 0 to x.
- If $x \geq 3$, we have passed the whole density, so $F_X(x) = 1$.

For $0 < x < 3$:

$$\begin{aligned} F_X(x) &= \int_0^x \frac{1}{9}t^2\,dt \\ &= \frac{1}{9}\left[\frac{t^3}{3}\right]_0^x \\ &= \frac{1}{9}\left(\frac{x^3}{3} - 0\right) \\ &= \frac{x^3}{27}. \end{aligned}$$

So:

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \dfrac{x^3}{27}, & 0 < x < 3, \\ 1, & x \geq 3. \end{cases}$$

Part c: Calculate $P(X > 1)$.

The CDF gives $P(X \leq 1)$, so use:

$$P(X > 1) = 1 - P(X \leq 1) = 1 - F_X(1).$$

Now insert $x = 1$:

$$F_X(1) = \frac{1^3}{27} = \frac{1}{27}.$$

Therefore:

$$\begin{aligned} P(X > 1) &= 1 - \frac{1}{27} \\ &= \frac{27}{27} - \frac{1}{27} \\ &= \frac{26}{27}. \end{aligned}$$

Part d: Expected value and standard deviation.

First calculate $E(X)$:

$$\begin{aligned} E(X) &= \int_0^3 x f(x)\,dx \\ &= \int_0^3 x \cdot \frac{1}{9}x^2\,dx \\ &= \frac{1}{9}\int_0^3 x^3\,dx \\ &= \frac{1}{9}\left[\frac{x^4}{4}\right]_0^3 \\ &= \frac{1}{9}\cdot\frac{3^4}{4} \\ &= \frac{1}{9}\cdot\frac{81}{4} \\ &= \frac{9}{4}. \end{aligned}$$

So the average value is $9/4 = 2.25$.

Now calculate $E(X^2)$, because we need it for the variance:

$$\begin{aligned} E(X^2) &= \int_0^3 x^2 f(x)\,dx \\ &= \int_0^3 x^2 \cdot \frac{1}{9}x^2\,dx \\ &= \frac{1}{9}\int_0^3 x^4\,dx \\ &= \frac{1}{9}\left[\frac{x^5}{5}\right]_0^3 \\ &= \frac{1}{9}\cdot\frac{3^5}{5} \\ &= \frac{1}{9}\cdot\frac{243}{5} \\ &= \frac{27}{5}. \end{aligned}$$

Now variance:

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{27}{5} - \left(\frac{9}{4}\right)^2 \\ &= \frac{27}{5} - \frac{81}{16}. \end{aligned}$$

Use common denominator 80:

$$\begin{aligned} V(X) &= \frac{432}{80} - \frac{405}{80} \\ &= \frac{27}{80}. \end{aligned}$$

Standard deviation:

$$\begin{aligned} D(X) &= \sqrt{V(X)} \\ &= \sqrt{\frac{27}{80}} \\ &\approx 0.581. \end{aligned}$$

Answer

$$\boxed{a = \frac{1}{9}}$$

$$\boxed{F_X(x) = \begin{cases} 0, & x \leq 0, \\ \frac{x^3}{27}, & 0 < x < 3, \\ 1, & x \geq 3 \end{cases}}$$

$$\boxed{P(X > 1) = \frac{26}{27} \approx 0.963}$$

$$\boxed{E(X) = \frac{9}{4} = 2.25}, \qquad \boxed{D(X) = \sqrt{\frac{27}{80}} \approx 0.581.}$$

What to memorize for the exam

For density questions:

- First make the total area equal to 1.
- Then build the CDF by integrating the density.
- For $P(X > c)$, use $1 - F(c)$.
- For standard deviation, calculate variance first.

Common trap

Do not forget the CDF pieces outside the interval. A CDF is not only $\frac{x^3}{27}$. It must also be 0 before the interval and 1 after the interval.

Repeated exam pattern

This is the same basic pattern as 2026-01, Uppgift 2: continuous density normalization, CDF, probability, expected value, and standard deviation. The density function changed, but the workflow is the same.

Uppgift 3

What this question is really asking

We have two independent Poisson random variables:

$$X_1,\ X_2 \sim \text{Po}(2).$$

This means both variables count something, and the average count is 2.

We need to calculate:

- one exact Poisson probability,
- one interval Poisson probability,
- a probability involving the maximum of X_1 and X_2,
- expected value and variance for a linear combination.

How to recognize this type of question

The clues are:

- The notation $\text{Po}(\mu)$ means Poisson distribution.
- The question says $\mu = 2$, so the Poisson mean is 2.
- It says X_1 and X_2 are independent, so probabilities involving both variables can be multiplied.
- The word max means the largest of the two values.

Tiny theory needed

For a Poisson variable $X \sim \text{Po}(\mu)$, the probability of exactly k is:

$$P(X = k) = \frac{\mu^k}{k!}e^{-\mu}.$$

For Poisson:

$$E(X) = \mu, \qquad V(X) = \mu.$$

So for $\mu = 2$:

$$E(X_1) = E(X_2) = 2, \qquad V(X_1) = V(X_2) = 2.$$

If X_1 and X_2 are independent, then:

$$P(X_1 \leq 2,\ X_2 \leq 2) = P(X_1 \leq 2)P(X_2 \leq 2).$$

Formula and what everything means

Poisson formula:

$$P(X = k) = \frac{\mu^k}{k!}e^{-\mu}.$$

Here:

- k is the exact number we want.
- μ is the average count.
- $k!$ means $k \cdot (k-1) \cdots 1$. For example, $3! = 3 \cdot 2 \cdot 1 = 6$.
- e is the calculator number 2.71828....

For variance of a linear combination:

$$V(aX + bY) = a^2V(X) + b^2V(Y)$$

when X and Y are independent. The coefficients are squared, so negative signs disappear in variance.

Step-by-step solution

Part a: Calculate $P(X_1 = 1)$.

Here $k = 1$ and $\mu = 2$:

$$\begin{aligned} P(X_1 = 1) &= \frac{2^1}{1!}e^{-2} \\ &= 2e^{-2} \\ &\approx 0.271. \end{aligned}$$

Now calculate $P(1 \leq X_1 < 4)$. Since X_1 is a count, this means:

$$X_1 = 1,\ 2,\ \text{or } 3.$$

So:

$$P(1 \leq X_1 < 4) = P(X_1 = 1) + P(X_1 = 2) + P(X_1 = 3).$$

Calculate each part:

$$\begin{aligned} P(X_1 = 1) &= \frac{2^1}{1!}e^{-2} = 2e^{-2}, \\ P(X_1 = 2) &= \frac{2^2}{2!}e^{-2} = \frac{4}{2}e^{-2} = 2e^{-2}, \\ P(X_1 = 3) &= \frac{2^3}{3!}e^{-2} = \frac{8}{6}e^{-2} = \frac{4}{3}e^{-2}. \end{aligned}$$

Add them:

$$\begin{aligned} P(1 \leq X_1 < 4) &= \left(2 + 2 + \frac{4}{3}\right)e^{-2} \\ &= \frac{16}{3}e^{-2} \\ &\approx 0.722. \end{aligned}$$

Part b: Calculate $P(\max(X_1, X_2) \leq 2)$.

The maximum is at most 2 only if both variables are at most 2. So:

$$\max(X_1, X_2) \leq 2$$

means:

$$X_1 \leq 2 \text{ and } X_2 \leq 2.$$

Because X_1 and X_2 are independent:

$$P(\max(X_1, X_2) \leq 2) = P(X_1 \leq 2)P(X_2 \leq 2).$$

They have the same distribution, so these two probabilities are equal:

$$P(\max(X_1, X_2) \leq 2) = [P(X_1 \leq 2)]^2.$$

Now calculate $P(X_1 \leq 2)$:

$$\begin{aligned} P(X_1 \leq 2) &= P(X_1 = 0) + P(X_1 = 1) + P(X_1 = 2) \\ &= \frac{2^0}{0!}e^{-2} + \frac{2^1}{1!}e^{-2} + \frac{2^2}{2!}e^{-2} \\ &= (1 + 2 + 2)e^{-2} \\ &= 5e^{-2}. \end{aligned}$$

Therefore:

$$\begin{aligned} P(\max(X_1, X_2) \leq 2) &= (5e^{-2})^2 \\ &= 25e^{-4} \\ &\approx 0.458. \end{aligned}$$

Part c: Calculate $E(Y)$ and $V(Y)$.

Using the intended reading $Y = X_2 - 2X_1$:

$$Y = X_2 - 2X_1.$$

Expected value is linear. That means we can move constants outside:

$$\begin{aligned} E(Y) &= E(X_2 - 2X_1) \\ &= E(X_2) - 2E(X_1). \end{aligned}$$

Since both means are 2:

$$\begin{aligned} E(Y) &= 2 - 2 \cdot 2 \\ &= 2 - 4 \\ &= -2. \end{aligned}$$

For variance, we square the coefficients:

$$\begin{aligned} V(Y) &= V(X_2 - 2X_1) \\ &= V(X_2) + (-2)^2 V(X_1). \end{aligned}$$

The minus sign disappears because $(-2)^2 = 4$:

$$\begin{aligned} V(Y) &= 2 + 4 \cdot 2 \\ &= 2 + 8 \\ &= 10. \end{aligned}$$

Answer

$$\boxed{P(X_1 = 1) = 2e^{-2} \approx 0.271}$$

$$\boxed{P(1 \leq X_1 < 4) = \frac{16}{3}e^{-2} \approx 0.722}$$

$$\boxed{P(\max(X_1, X_2) \leq 2) = 25e^{-4} \approx 0.458}$$

$$\boxed{E(Y) = -2, \qquad V(Y) = 10}$$

for the intended reading $Y = X_2 - 2X_1$.

What to memorize for the exam

For Poisson:

$$P(X = k) = \frac{\mu^k}{k!}e^{-\mu}, \qquad E(X) = \mu, \qquad V(X) = \mu.$$

For a maximum:

$$\max(X_1, X_2) \leq c$$

means both variables are $\leq c$.

Common trap

Variance can never be negative. When calculating $V(X_2 - 2X_1)$, the -2 must be squared:

$$(-2)^2 = 4.$$

So the variance contribution is $4V(X_1)$, not $-2V(X_1)$.

Repeated exam pattern

The Poisson parts are a new pattern in the processed folders so far. The linear-combination variance idea is related to 2026-01, Uppgift 4b: coefficients must be squared when working with variance.

Uppgift 4

What this question is really asking

This is a maximum likelihood question, usually shortened to ML.

We are given a density:

$$f_X(x) = \theta x^{\theta - 1}, \qquad 0 < x < 1.$$

The unknown parameter is θ. The question wants the ML estimate of θ, meaning the value of θ that makes the observed sample most likely.

How to recognize this type of question

The clue is the phrase:

"Ange ML-skattningen av θ".

That tells us immediately:

- write the likelihood,
- take the logarithm,
- differentiate,
- set the derivative equal to zero,
- solve for θ.

Tiny theory needed

The likelihood is the product of the density values at all observed data points:

$$L(\theta) = \prod_{i=1}^{n} f_X(x_i).$$

The product sign $\prod$ just means multiply:

$$\prod_{i=1}^{n} f_X(x_i) = f_X(x_1) f_X(x_2) \cdots f_X(x_n).$$

We take logarithms because products are annoying, but logs turn products into sums. This makes differentiation much easier.

Formula and what everything means

The density is:

$$f_X(x_i) = \theta x_i^{\theta-1}.$$

Here:

- x_i is one observed value in the sample.
- n is the number of observations.
- θ is the unknown number we are estimating.
- $L(\theta)$ is the likelihood, a function of θ.

ML recipe:

$$\ell(\theta) = \ln L(\theta),$$

then solve:

$$\ell'(\theta) = 0.$$

Step-by-step solution

Start with the likelihood:

$$L(\theta) = \prod_{i=1}^{n} \theta x_i^{\theta-1}.$$

There is one θ for every observation, so multiplying them gives θ^n:

$$L(\theta) = \theta^n \prod_{i=1}^{n} x_i^{\theta-1}.$$

Now take the natural log:

$$\begin{aligned} \ell(\theta) &= \ln L(\theta) \\ &= \ln\left(\theta^n \prod_{i=1}^{n} x_i^{\theta-1}\right). \end{aligned}$$

Use log rules:

$$\ell(\theta) = n \ln \theta + \sum_{i=1}^{n} \ln\left(x_i^{\theta-1}\right).$$

Another log rule says $\ln(a^b) = b\ln(a)$:

$$\begin{aligned} \ell(\theta) &= n \ln \theta + \sum_{i=1}^{n} (\theta - 1) \ln x_i \\ &= n \ln \theta + (\theta - 1) \sum_{i=1}^{n} \ln x_i. \end{aligned}$$

Now differentiate with respect to θ:

$$\ell'(\theta) = \frac{n}{\theta} + \sum_{i=1}^{n} \ln x_i.$$

Set the derivative equal to zero:

$$\frac{n}{\theta} + \sum_{i=1}^{n} \ln x_i = 0.$$

Move the sum to the other side:

$$\frac{n}{\theta} = -\sum_{i=1}^{n} \ln x_i.$$

Now solve for θ:

$$\begin{aligned} \theta &= \frac{n}{-\sum_{i=1}^{n} \ln x_i} \\ &= -\frac{n}{\sum_{i=1}^{n} \ln x_i}. \end{aligned}$$

Since every x_i is between 0 and 1, every $\ln x_i$ is negative. So the denominator is negative, and the minus sign makes the estimate positive.

Answer

$$\boxed{\hat{\theta}_{ML} = -\frac{n}{\sum_{i=1}^{n} \ln x_i}}$$

What to memorize for the exam

For ML questions:

$$\text{likelihood} \rightarrow \text{log-likelihood} \rightarrow \text{differentiate} \rightarrow \text{set equal to zero.}$$

That is the exam routine almost every time.

Common trap

Do not stop after writing the likelihood. Most points come from taking the log, differentiating, and solving for the parameter.

Repeated exam pattern

This is the same broad ML pattern as 2026-01, Uppgift 5. The distribution is different, but the solving recipe is the same: build L, take $\ln L$, differentiate, solve.

Uppgift 5

What this question is really asking

We have 8 measurements that are assumed to come from a normal distribution:

$$N(\mu, \sigma).$$

The question asks about σ^2 and σ.

Here:

- σ^2 is the variance, which measures spread.
- σ is the standard deviation, which is the square root of the variance.

Part a asks for point estimates, meaning one best guess number for each.

Part b asks for a 90% confidence interval for σ, meaning an interval of plausible values for the true standard deviation.

How to recognize this type of question

The clues are:

- The data are assumed normal.
- The question asks about σ, not μ.
- A confidence interval for σ uses the chi-square distribution, not the normal distribution.

Tiny theory needed

For sample variance, we use:

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right).$$

This formula uses:

- $\sum x_i$, the sum of the observations,
- $\sum x_i^2$, the sum of squared observations,
- $\bar{x}$, the sample mean,
- $n-1$, because this is the usual unbiased estimate of variance.

For a confidence interval for σ, we first make an interval for σ^2, then take square roots.

Formula and what everything means

Sample mean:

$$\bar{x} = \frac{\sum x_i}{n}.$$

Here $\bar{x}$ is the average of the observed data.

Sample variance:

$$s^2 = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}.$$

Here s^2 estimates σ^2.

Confidence interval for σ^2:

$$\left(\frac{(n-1)s^2}{\chi^2_{0.95,n-1}}, \frac{(n-1)s^2}{\chi^2_{0.05,n-1}}\right).$$

For a 90% interval, we put 5% in each tail, so we use 0.95 and 0.05 chi-square values.

Step-by-step solution

We are given:

$$n = 8, \qquad \sum_{i=1}^{8} x_i = 114.1, \qquad \sum_{i=1}^{8} x_i^2 = 1629.45.$$

Part a: Point estimate of σ^2 and σ.

First calculate the sample mean:

$$\begin{aligned}\bar{x} &= \frac{114.1}{8} \\ &= 14.2625.\end{aligned}$$

Now calculate the numerator of the variance:

$$\sum x_i^2 - n\bar{x}^2 = 1629.45 - 8(14.2625)^2.$$

Calculate the square:

$$(14.2625)^2 = 203.41890625.$$

Multiply by 8:

$$8(14.2625)^2 = 1627.35125.$$

Subtract:

$$1629.45 - 1627.35125 = 2.09875.$$

Now divide by $n - 1 = 7$:

$$\begin{aligned} s^2 &= \frac{2.09875}{7} \\ &\approx 0.2998. \end{aligned}$$

So:

$$\hat{\sigma}^2 = s^2 \approx 0.2998.$$

Now take the square root to estimate σ:

$$\begin{aligned} s &= \sqrt{0.2998} \\ &\approx 0.5476. \end{aligned}$$

Part b: 90% confidence interval for σ.

We use $n - 1 = 7$ degrees of freedom.

From a chi-square table:

$$\chi^2_{0.95,7} \approx 14.067, \qquad \chi^2_{0.05,7} \approx 2.167.$$

Also:

$$(n-1)s^2 = 7 \cdot 0.2998 = 2.09875.$$

So the interval for σ^2 is:

$$\left(\frac{2.09875}{14.067}, \frac{2.09875}{2.167}\right) \approx (0.1492,\ 0.9683).$$

But the question asks for σ, not σ^2. So take square roots of both endpoints:

$$\begin{aligned} \sigma &\in \left(\sqrt{0.1492}, \sqrt{0.9683}\right) \\ &\approx (0.386,\ 0.984). \end{aligned}$$

Answer

$$\boxed{\hat{\sigma}^2 = s^2 \approx 0.2998}$$

$$\boxed{\hat{\sigma} = s \approx 0.5476}$$

$$\boxed{\text{90\% confidence interval for } \sigma: \ (0.386,\ 0.984)}$$

What to memorize for the exam

If the question asks for an interval for σ or σ^2 from normal data, think:

chi-square distribution.

Do not use a z-interval or t-interval for σ.

Common trap

A very common mistake is to calculate a confidence interval for the mean μ. But this question asks for σ, the spread. That is a different formula.

Repeated exam pattern

No exact earlier matching pattern found yet. Track this as a new pattern: point estimate and chi-square confidence interval for normal standard deviation.

Uppgift 6

What this question is really asking

This question is about testing whether the average passenger weight has increased.

Old belief:

$$\mu = 80.$$

Now we want to test whether:

$$\mu > 80.$$

We know the population standard deviation:

$$\sigma = 10.$$

Because σ is known, this is a z-test.

How to recognize this type of question

The clues are:

- It says "testa", so this is a hypothesis test.
- It asks whether the mean has increased, so the alternative is $>$.
- It gives $\sigma = 10$, so use a normal z-test instead of a t-test.
- Part c asks how large n must be to detect an increase with 80% probability. That is a power/sample-size question.

Tiny theory needed

A hypothesis test compares two ideas:

- H_0: the default idea, usually "no increase".
- H_1: the thing we are looking for evidence of.

Here we are looking for evidence that the mean is larger than 80, so:

$$H_0 : \mu = 80, \qquad H_1 : \mu > 80.$$

The test statistic tells us how many standard errors the observed mean is above 80.

Formula and what everything means

The z-test statistic is:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

Here:

- $\bar{x}$ is the sample average.
- μ_0 is the mean in the null hypothesis, here 80.
- σ is the known standard deviation, here 10.
- n is the sample size.
- $\sigma/\sqrt{n}$ is the standard error, meaning the typical spread of the sample average.

For a one-sided 5% test:

$$z_{0.95} \approx 1.645.$$

We reject H_0 if the test statistic is bigger than 1.645.

Step-by-step solution

Part a: Set up the hypotheses.

The old mean is 80, and we are checking if the mean has increased. So:

$$H_0 : \mu = 80$$

and

$$H_1 : \mu > 80.$$

The significance level is 5%. That means we allow a 5% risk of falsely saying there is an increase when there is not.

Part b: Perform the test for $n = 100$ and $\bar{x} = 83$.

Insert the values:

$$\bar{x} = 83, \qquad \mu_0 = 80, \qquad \sigma = 10, \qquad n = 100.$$

First calculate the standard error:

$$\begin{aligned} \frac{\sigma}{\sqrt{n}} &= \frac{10}{\sqrt{100}} \\ &= \frac{10}{10} \\ &= 1. \end{aligned}$$

Now calculate the z-value:

$$\begin{aligned} z &= \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \\ &= \frac{83 - 80}{1} \\ &= 3. \end{aligned}$$

Compare with the critical value:

$$1.645.$$

Since:

$$3 > 1.645,$$

we reject H_0.

In plain language: the sample mean 83 is far enough above 80 that we say there is evidence the average weight has increased.

Part c: Find n for 80% probability of detecting $\mu = 82$.

The test rejects when:

$$\bar{X} > 80 + 1.645\frac{10}{\sqrt{n}}.$$

This is the rejection boundary. The sample mean must be above this number to reject H_0.

Now assume the true mean is actually:

$$\mu = 82.$$

Then:

$$\bar{X} \sim N\left(82, \frac{10}{\sqrt{n}}\right).$$

We want:

$$P(\text{reject } H_0) > 0.80.$$

Using the rejection rule:

$$P\left(\bar{X} > 80 + 1.645\frac{10}{\sqrt{n}}\right) > 0.80.$$

Standardize this probability. Subtract the true mean 82, then divide by the standard error $10/\sqrt{n}$:

$$P\left(Z > \frac{80 + 1.645\frac{10}{\sqrt{n}} - 82}{10/\sqrt{n}}\right) > 0.80.$$

The problem tells us:

$$\Phi(0.84) \approx 0.80.$$

That means:

$$P(Z > -0.84) \approx 0.80.$$

So we need the cutoff inside the probability to be at most -0.84:

$$\frac{80 + 1.645\frac{10}{\sqrt{n}} - 82}{10/\sqrt{n}} \leq -0.84.$$

Simplify the top:

$$80 - 82 = -2.$$

So:

$$\frac{-2 + 1.645\frac{10}{\sqrt{n}}}{10/\sqrt{n}} \leq -0.84.$$

Split the fraction into two parts:

$$\frac{-2}{10/\sqrt{n}} + 1.645 \leq -0.84.$$

Since dividing by $10/\sqrt{n}$ is the same as multiplying by $\sqrt{n}/10$:

$$-\frac{2\sqrt{n}}{10} + 1.645 \leq -0.84$$
$$-0.2\sqrt{n} + 1.645 \leq -0.84.$$

Move 1.645 to the right:

$$-0.2\sqrt{n} \leq -0.84 - 1.645$$
$$-0.2\sqrt{n} \leq -2.485.$$

Divide by -0.2. Because we divide by a negative number, the inequality sign flips:

$$\sqrt{n} \geq 12.425.$$

Square both sides:

$$n \geq (12.425)^2$$
$$n \geq 154.38.$$

The sample size must be a whole number, and it must be large enough, so round up:

$$n = 155.$$

Answer

$$\boxed{H_0 : \mu = 80, \qquad H_1 : \mu > 80}$$

$$\boxed{z = 3}$$

$$\boxed{\text{Reject } H_0 \text{ at the 5\% level. There is evidence that the mean has increased.}}$$

$$\boxed{n = 155 \text{ is needed for about 80\% power when } \mu = 82.}$$

What to memorize for the exam

If the population standard deviation σ is known, use:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

If the question says "has increased", use a right-tailed test:

$$H_1 : \mu > \mu_0.$$

Common trap

In part c, do not plug in $\bar{x} = 83$. Part c is not about the old sample anymore. It asks how large n must be if the true mean is 82.

Repeated exam pattern

This is related to 2026-01, Uppgift 6 because both are hypothesis tests for a mean. The difference is that this question has known σ and a one-sided alternative, so it uses a z-test instead of a t-test.

Mini-cheatsheet from this exam

Probability

Complement:

$$P(A) = 1 - P(A^c).$$

Conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Union:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Continuous density

To make a density:

$$\int f(x)\,dx = 1.$$

CDF:

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt.$$

Right-tail probability:

$$P(X > c) = 1 - F(c).$$

Variance:

$$V(X) = E(X^2) - [E(X)]^2.$$

Poisson

If $X \sim \mathrm{Po}(\mu)$:

$$P(X = k) = \frac{\mu^k}{k!} e^{-\mu}.$$

$$E(X) = \mu, \qquad V(X) = \mu.$$

For independent variables:

$$P(X_1 \le c,\ X_2 \le c) = P(X_1 \le c) P(X_2 \le c).$$

Maximum likelihood

Standard ML workflow:

$$L(\theta) \to \ln L(\theta) \to \text{differentiate} \to \text{set equal to zero.}$$

Variance and standard deviation

Sample mean:

$$\bar{x} = \frac{\sum x_i}{n}.$$

Sample variance:

$$s^2 = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}.$$

Standard deviation:

$$s = \sqrt{s^2}.$$

Confidence interval for σ

For normal data:

$$\left(\sqrt{\frac{(n-1)s^2}{\chi^2_{0.95,n-1}}}, \sqrt{\frac{(n-1)s^2}{\chi^2_{0.05,n-1}}}\right)$$

for a 90% interval.

Hypothesis test for mean, known σ

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

Right-tailed 5% test:

$$\text{reject if } z > 1.645.$$

Power/sample size trick

First write the rejection boundary, then calculate the probability of crossing that boundary under the new true mean.